

ShopEZ : E-Commerce Application

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INTRODUCTION

ShopEZ is your one-stop destination for effortless online shopping. With a user-friendly interface and a comprehensive product catalog, finding the perfect items has never been easier. Seamlessly navigate through detailed product descriptions, customer reviews, and available discounts to make informed decisions. Enjoy a secure checkout process and receive instant order confirmation. For sellers, our robust dashboard provides efficient order management and insightful analytics to drive business growth. Experience the future of online shopping with ShopEZ today.

Seamless Checkout Process

Effortless Product Discovery

Personalized Shopping Experience

Efficient Order Management for Sellers

Insightful Analytics for Business Growth

SCENARIO:

Sarah's Birthday Gift

Sarah, a busy professional, is scrambling to find the perfect birthday gift for her best friend, Emily. She knows Emily loves fashion accessories, but with her hectic schedule, she hasn't had time to browse through multiple websites to find the ideal present. Feeling overwhelmed, Sarah turns to ShopEZ to simplify her search.

1. Effortless Product Discovery : Sarah opens ShopEZ and navigates to the fashion accessories category. She's greeted with a diverse range of options, from chic handbags to elegant jewelry. Using the filtering options, Sarah selects "bracelets" and refines her search based on Emily's preferred style and budget.

2. Personalized Recommendations: As Sarah scrolls through the curated selection of bracelets, she notices a section labeled "Recommended for You." Intrigued, she clicks on it and discovers a stunning gold bangle that perfectly matches Emily's taste. Impressed by the personalized recommendation, Sarah adds it to her cart.

3. Seamless Checkout Process: With the bracelet in her cart, Sarah proceeds to checkout. She enters Emily's address as the shipping destination and selects her preferred payment method. Thanks to ShopEZ's secure and efficient checkout process, Sarah completes the transaction in just a few clicks.

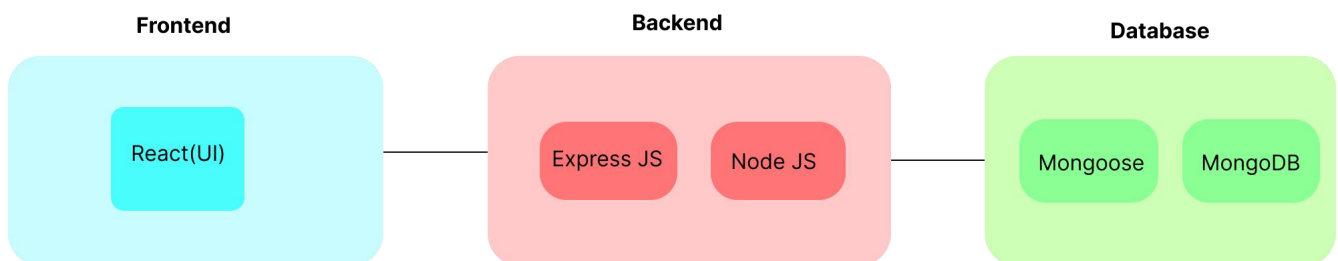
4. Order Confirmation: Moments after placing her order, Sarah receives a confirmation email from ShopEZ. Relieved to have found the perfect gift for Emily, she eagerly awaits its arrival.

5. Efficient Order Management for Sellers: Meanwhile, on the other end, the seller of the gold bangle receives a notification of Sarah's purchase through ShopEZ's seller dashboard. They quickly process the order and prepare it for shipment, confident in ShopEZ's streamlined order management system.

6. Celebrating with Confidence : On Emily's birthday, Sarah presents her with the beautifully packaged bracelet, knowing it was chosen with care and thoughtfulness. Emily's eyes light up with joy as she adorns the bracelet, grateful for Sarah's thoughtful gesture.

In this scenario, ShopEZ proves to be the perfect solution for Sarah's busy lifestyle, offering a seamless and personalized shopping experience. From effortless product discovery to secure checkout and efficient order management, ShopEZ simplifies the entire process, allowing Sarah to celebrate Emily's birthday with confidence and ease.

TECHNICAL ARCHITECTURE:

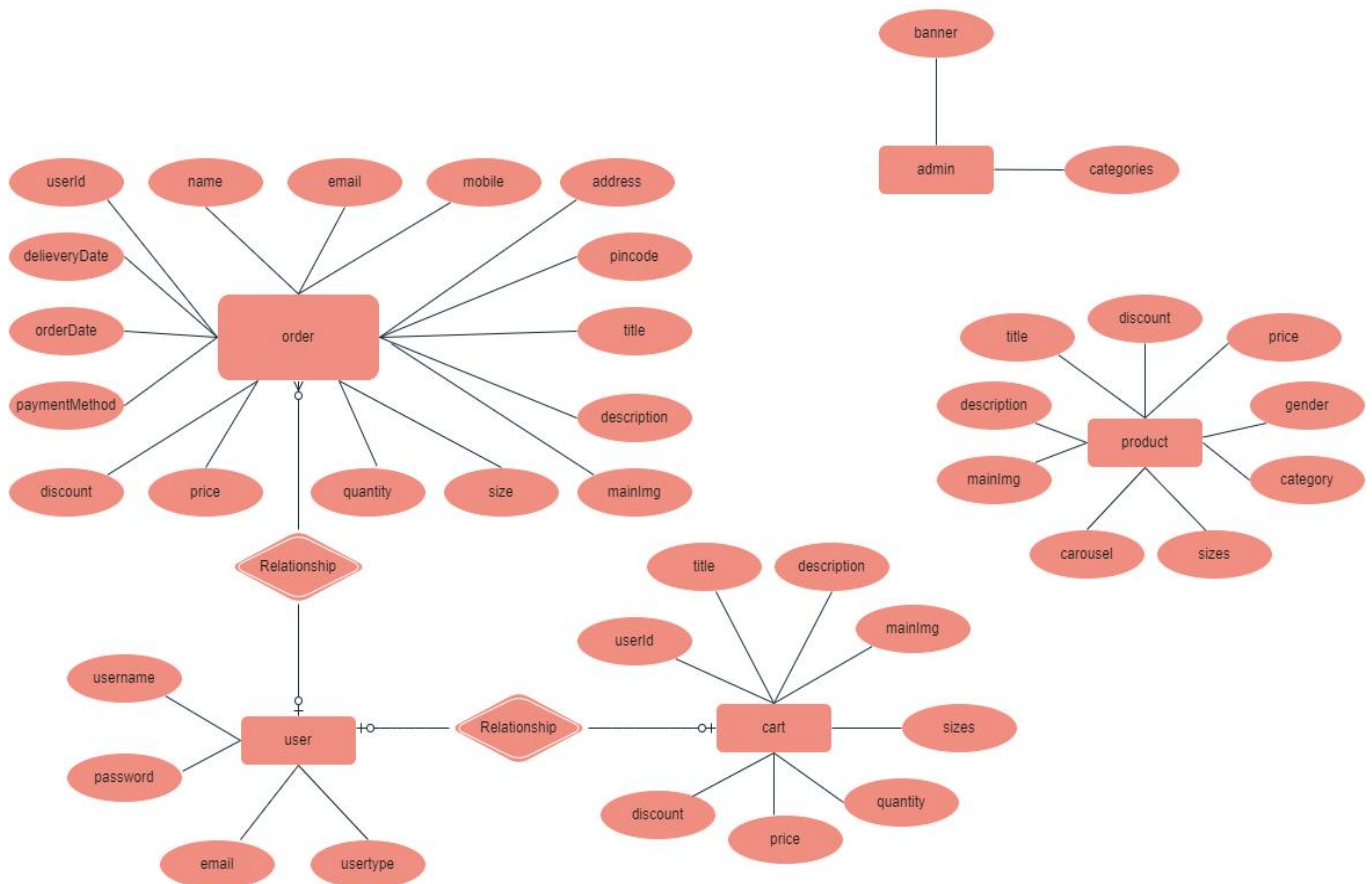


In this architecture diagram:

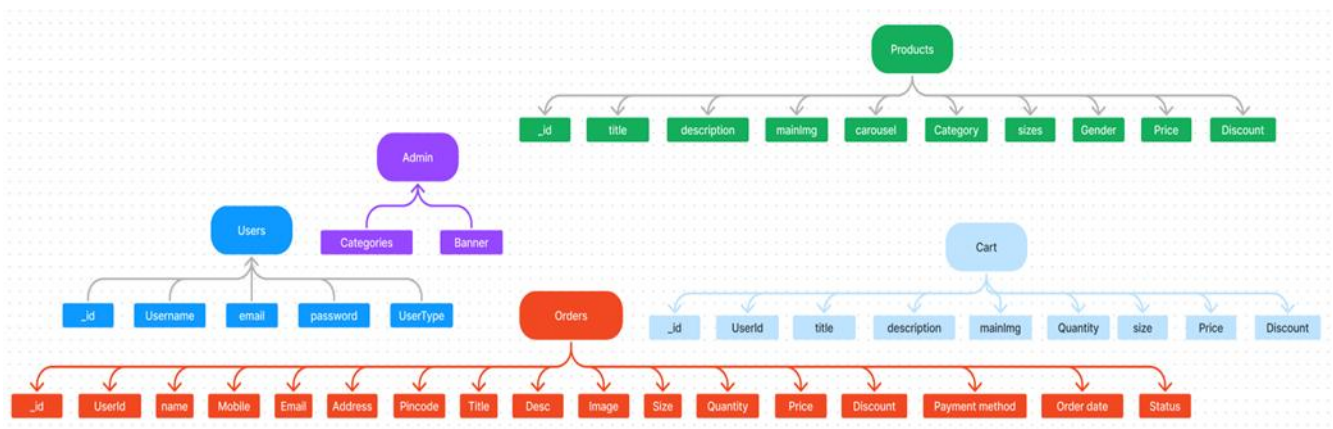
- The frontend is represented by the "Frontend" section, including user interface components such as User Authentication, Cart, Products, Profile, Admin dashboard, etc.,
- The backend is represented by the "Backend" section, consisting of API endpoints for Users, Orders, Products, etc., It also includes Admin Authentication and an Admin Dashboard.
- The Database section represents the database that stores collections for Users, cart, Orders and Product.

ER DIAGRAM:

ER-MODEL



- The Database section represents the database that stores collections for Users, Admin, Cart, Orders and products.



The ShopEZ ER-diagram represents the entities and relationships involved in an e-commerce system. It illustrates how users, products, cart, and orders are interconnected. Here is a breakdown of the entities and their relationships:

USER: Represents the individuals or entities who are registered in the platform.

Admin: Represents a collection with important details such as Banner image and Categories.

Products: Represents a collection of all the products available in the platform.

Cart: This collection stores all the products that are added to the cart by users. Here, the elements in the cart are differentiated by the user Id.

Orders: This collection stores all the orders that are made by the users in the platform.

FEATURES:

1. **Comprehensive Product Catalog:** ShopEZ boasts an extensive catalog of products, offering a diverse range of items and options for shoppers. You can effortlessly explore and discover various products, complete with detailed descriptions, customer reviews, pricing, and available discounts, to find the perfect items for your needs.
2. **Shop Now Button:** Each product listing features a convenient "Shop Now" button. When you find a product that aligns with your preferences, simply click on the button to initiate the purchasing process.
3. **Order Details Page:** Upon clicking the "Shop Now" button, you will be directed to an order details page. Here, you can provide relevant information such as your shipping address, preferred payment method, and any specific product requirements.
4. **Secure and Efficient Checkout Process:** ShopEZ guarantees a secure and efficient checkout process. Your personal information will be handled with the utmost security, and we strive to make the purchasing process as swift and trouble-free as possible.
5. **Order Confirmation and Details:** After successfully placing an order, you will receive a confirmation notification. Subsequently, you will be directed to an order details page, where you can review all pertinent information about your order, including shipping details, payment method, and any specific product requests you specified.

In addition to these user-centric features, ShopEZ provides a robust seller dashboard, offering sellers an array of functionalities to efficiently manage their products and sales. With the seller dashboard, sellers can add and oversee multiple product listings, view order history, monitor customer activity, and access order details for all purchases.

ShopEZ is designed to elevate your online shopping experience by providing a seamless and user-friendly way to discover and purchase products. With our efficient checkout process, comprehensive product catalog, and robust seller dashboard, we ensure a convenient and enjoyable online shopping experience for both shoppers and sellers alike.

PREREQUISITES:

To develop a full-stack e-commerce app using React JS, Node.js, and MongoDB, there are several prerequisites you should consider. Here are the key prerequisites for developing such an application:

Node.js and npm:

Install Node.js, which includes npm (Node Package Manager), on your development machine. Node.js is required to run JavaScript on the server side.

- Download: <https://nodejs.org/en/download/>
- Installation instructions: <https://nodejs.org/en/download/package-manager/>

MongoDB:

MongoDB is a flexible and scalable NoSQL database that stores data in a JSON-like format. It provides high performance, horizontal scalability, and seamless integration with Node.js, making it ideal for handling large amounts of structured and unstructured data.

Set up a MongoDB database to store your application's data.

- Download: <https://www.mongodb.com/try/download/community>
- Installation instructions: <https://docs.mongodb.com/manual/installation/>

Express.js:

Express.js is a fast and minimalist web application framework for Node.js. It simplifies the process of creating robust APIs and web applications, offering features like routing, middleware support, and modular architecture.

Install Express.js, a web application framework for Node.js, which handles server-side routing, middleware, and API development.

- Installation: Open your command prompt or terminal and run the following command: **npm install express**

React.js:

React.js is a popular JavaScript library for building user interfaces. It enables developers to create interactive and reusable UI components, making it easier to build dynamic and responsive web applications.

Install React.js, a JavaScript library for building user interfaces.

<https://reactjs.org/docs/create-a-new-react-app.html>

HTML, CSS, and JavaScript: Basic knowledge of HTML for creating the structure of your app, CSS for styling, and JavaScript for client-side interactivity is essential.

Database Connectivity: Use a MongoDB driver or an Object-Document Mapping (ODM) library like Mongoose to connect your Node.js server with the MongoDB database and perform CRUD (Create, Read, Update, Delete) operations.

Front-end Framework: Utilize React Js to build the user-facing part of the application, including entering booking room, status of the booking, and user interfaces for the admin dashboard. For making better UI we have also used some libraries like material UI and bootstrap

Version Control: Use Git for version control, enabling collaboration and tracking changes throughout the development process. Platforms like GitHub or Bitbucket can host your repository.

Git: Download and installation instructions can be found at:

<https://git-scm.com/downloads> [scm.com/downloads](https://git-scm.com/downloads)

Development Environment: Choose a code editor or Integrated Development Environment (IDE) that suits your preferences, such as Visual Studio Code, Sublime Text, or WebStorm.

- Visual Studio Code: Download from <https://code.visualstudio.com/download>

Install Dependencies:

- Navigate into the cloned repository directory:
cd ShopEZ—e-commerce-App-MERN
- Install the required dependencies by running the following command:
npm install

Start the Development Server:

- To start the development server, execute the following command:
npm start
- This app will be accessible at <http://localhost:3000>

You have successfully installed and set up the SB application on your local machine. You can now proceed with further customization, development, and testing as needed.

PROJECT FLOW:-

Before starting to work on this project, let's see the demo.

Demo link:-

https://drive.google.com/file/d/1Q0q8CSI5AnsXR7tzt55K35pyv67VR82/view?usp=drive_link

Use the code in:-

<https://drive.google.com/drive/folders/1L3Fqd1qydUTioetkTy6P1aLwEvxcDmCG?usp=sharing>

USER & ADMIN FLOW:

1. User Flow:

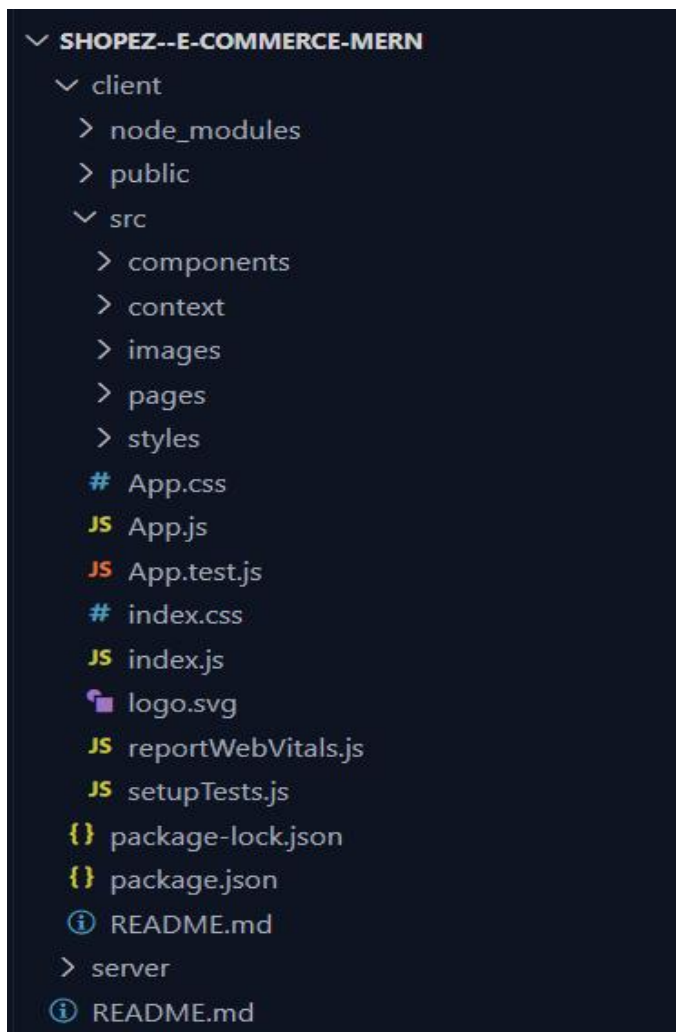
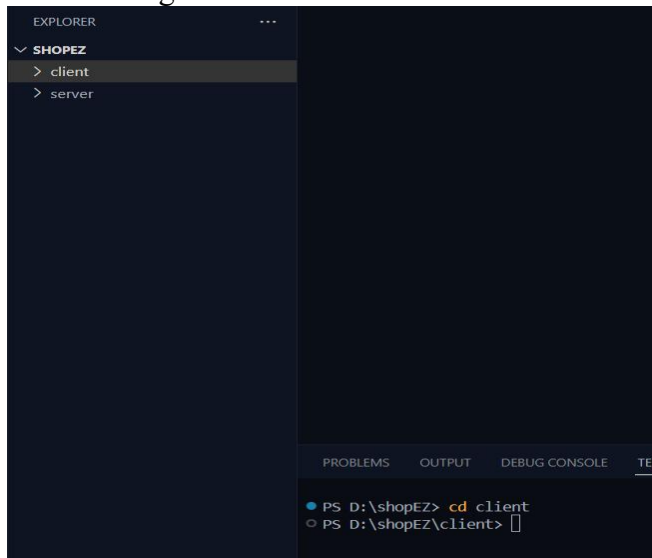
- Users start by registering for an account.
- After registration, they can log in with their credentials.
- Once logged in, they can check for the available products in the platform. • Users can add the products they wish to their carts and order.
- They can then proceed by entering address and payment details. • After ordering, they can check them in the profile section.

2. Admin Flow:

- Admins start by logging in with their credentials.
- Once logged in, they are directed to the Admin Dashboard.
- Admins can access the users list, products, orders, etc.,

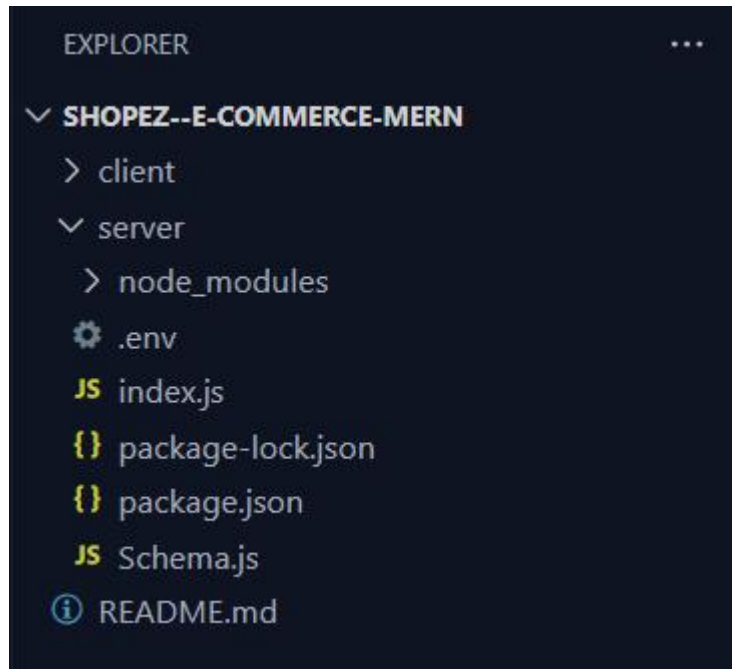
PROJECT STRUCTURE:

Referral Image



This structure assumes a React app and follows a modular approach. Here's a brief explanation of the main directories and files:

- src/components: Contains components related to the application such as, register, login, home, etc.,
- src/pages has the files for all the pages in the application.



MILESTONE 1 : PROJECT SETUP AND CONFIGURATION:

Install required tools and software:

- Node.js.
- MongoDB.
- Create-react-app.

Create project folders and files:

- Client folders.
- Server folders.

Install Packages:

Frontend npm Packages

- Node
- React
- Bootstrap.

Backend npm Packages

- Express.
- Mongoose.
- Body-parser
- Cors.

DATABASE DEVELOPMENT:

Reference Image:

```
server > JS index.js > ...
1  import express from "express";
2  import mongoose from "mongoose";
3  import cors from "cors";
4  import dotenv from "dotenv";
5
6  dotenv.config({ path: "./.env" });
7
8  const app = express();
9  app.use(express.json());
10 app.use(cors());
11
12 app.listen(3001, () => {
13   console.log("App server is running on port 3001");
14 });
15
16 const MongoUri = process.env.DRIVER_LINK;
17 const connectToMongo = async () => {
18   try {
19     await mongoose.connect(MongoUri);
20     console.log("Connected to your MongoDB database successfully");
21   } catch (error) {
22     console.log(error.message);
23   }
24 };
25
26 connectToMongo();
27
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\shopEZ> cd server
PS D:\shopEZ\server> node index.js
App server is running on port 3001
bad auth : authentication failed
PS D:\shopEZ\server> node index.js
App server is running on port 3001
Connected to your MongoDB database successfully
```

Schema use-case:

1. User Schema:

- Schema: userSchema
- Model: 'User'
- The User schema represents the user data and includes fields such as username, email, and password.
- It is used to store user information for registration and authentication purposes.
- The email field is marked as unique to ensure that each user has a unique email address

2. Product Schema:

- Schema: productSchema
- Model: 'Product'
- The Product schema represents the data of all the products in the platform.
- It is used to store information about the product details, which will later be useful for ordering .

3. Orders Schema:

- Schema: ordersSchema
- Model: 'Orders'
- The Orders schema represents the orders data and includes fields such as userId, product Id, product name, quantity, size, order date, etc.,
- It is used to store information about the orders made by users.
- The user Id field is a reference to the user who made the order.

4. Cart Schema:

- Schema: cartSchema
- Model: 'Cart'
- The Cart schema represents the cart data and includes fields such as userId, product Id, product name, quantity, size, order date, etc.,
- It is used to store information about the products added to the cart by users. • The user Id field is a reference to the user who has the product in cart.

5. Admin Schema:

- Schema: adminSchema
- Model: 'Admin'
- The admin schema has essential data such as categories, banner.

Code Explanation:

Schemas:

Now let us define the required schemas

```
JS Schema.js X
server > JS Schema.js > productSchema
1  import mongoose from "mongoose";
2
3  const userSchema = new mongoose.Schema({
4    username: {type: String},
5    password: {type: String},
6    email: {type: String},
7    usertype: {type: String}
8  });
9
10 const adminSchema = new mongoose.Schema({
11   banner: {type: String},
12   categories: {type: Array}
13 });
14
15 const productSchema = new mongoose.Schema({
16   title: {type: String},
17   description: {type: String},
18   mainImg: {type: String},
19   carousel: {type: Array},
20   sizes: {type: Array},
21   category: {type: String},
22   gender: {type: String},
23   price: {type: Number},
24   discount: {type: Number}
25 });
26
```

```
JS Schemajs X
server > JS Schemajs > [0] productSchema
27 const orderSchema = new mongoose.Schema({
28   userId: {type: String},
29   name: {type: String},
30   email: {type: String},
31   mobile: {type: String},
32   address: {type: String},
33   pincode: {type: String},
34   title: {type: String},
35   description: {type: String},
36   mainImg: {type: String},
37   size: {type: String},
38   quantity: {type: Number},
39   price: {type: Number},
40   discount: {type: Number},
41   paymentMethod: {type: String},
42   orderDate: {type: String},
43   deliveryDate: {type: String},
44   orderStatus: {type: String, default: 'order placed'}
45 })
46
47 const cartSchema = new mongoose.Schema({
48   userId: {type: String},
49   title: {type: String},
50   description: {type: String},
51   mainImg: {type: String},
52   size: {type: String},
53   quantity: {type: String},
54   price: {type: Number},
55   discount: {type: Number}
56 })
57
58
59 export const User = mongoose.model('users', userSchema);
60 export const Admin = mongoose.model('admin', adminSchema);
61 export const Product = mongoose.model('products', productSchema);
62 export const Orders = mongoose.model('orders', orderSchema);
63 export const Cart = mongoose.model('cart', cartSchema);
64
```

MILESTONE 2 : BACKEND DEVELOPMENT:

Setup express server:

- 1.Create index.js file in the server (backend folder).
- 2.Create a .env file and define port number to access it globally.
- 3.Configure the server by adding cors, body-parser.

Set Up Project Structure:

- Create a new directory for your project and set up a package.json file using the npm init command.
- Install necessary dependencies such as Express.js, Mongoose, and other required packages.

The screenshot shows the VS Code interface with the Explorer sidebar on the left displaying the project structure: SHOPEZ, client, server, node_modules, package-lock.json, and package.json. The main editor area shows the package.json file with the following content:

```
1 {
2   "name": "server",
3   "version": "1.0.0",
4   "description": "",
5   "main": "index.js",
6   "scripts": {
7     "test": "echo \\\"Error: no test specified\\\" && exit 1"
8   },
9   "keywords": [],
10  "author": "",
11  "license": "ISC",
12  "dependencies": {
13    "bcrypt": "^5.1.1",
14    "body-parser": "^1.20.2",
15    "cors": "^2.8.5",
16    "dotenv": "^16.4.5",
17    "express": "^4.19.1",
18    "mongoose": "^8.2.3"
19  }
20 }
```

The terminal at the bottom shows the output of the command `npm install express mongoose body-parser dotenv`:

```
PS D:\shopEZ\server> npm install express mongoose body-parser dotenv
added 85 packages, and audited 86 packages in 11s
14 packages are looking for funding
run `npm fund` for details
found 0 vulnerabilities
PS D:\shopEZ\server> npm i bcrypt cors
added 61 packages, and audited 147 packages in 9s
```

The screenshot shows the VS Code interface with the Explorer sidebar on the left displaying the project structure: SHOPEZ, client, server, node_modules, package-lock.json, package.json, and index.js. The main editor area shows the index.js file with the following content:

```
1 import express from "express";
2
3 const app = express();
4 app.use(express.json());
5
6 app.listen(3001, () => {
7   console.log("App server is running on port 3001");
8 });
9
```

The terminal at the bottom shows the output of the commands `cd server` and `node index.js`:

```
PS D:\shopEZ> cd server
PS D:\shopEZ\server> node index.js
App server is running on port 3001
```

2. Database Configuration:

- Set up a MongoDB database either locally or using a cloud-based MongoDB service like MongoDB Atlas or use locally with MongoDB compass.
- Create a database and define the necessary collections for admin, users, products, orders and other relevant data.

3. Create Express.js Server:

- Set up an Express.js server to handle HTTP requests and serve API endpoints.
- Configure middleware such as body-parser for parsing request bodies and cors for handling cross-origin requests.

4. Define API Routes:

- Create separate route files for different API functionalities such as users, orders, and authentication.
- Define the necessary routes for listing products, handling user registration and login, managing orders, etc.

- Implement route handlers using Express.js to handle requests and interact with the database.

5. Implement Data Models:

- Define Mongoose schemas for the different data entities like products, users, and orders.
- Create corresponding Mongoose models to interact with the MongoDB database.
- Implement CRUD operations (Create, Read, Update, Delete) for each model to perform database operations.

6. User Authentication:

- Create routes and middleware for user registration, login, and logout.
- Set up authentication middleware to protect routes that require user authentication.

7. Handle new products and Orders:

- Create routes and controllers to handle new product listings, including fetching products data from the database and sending it as a response.

- Implement ordering(buy) functionality by creating routes and controllers to handle order requests, including validation and database updates.

8. Admin Functionality:

- Implement routes and controllers specific to admin functionalities such as adding products, managing user orders, etc.
- Add necessary authentication and authorization checks to ensure only authorized admins can access these routes.

9. Error Handling:

- Implement error handling middleware to catch and handle any errors that occur during the API requests.
- Return appropriate error responses with relevant error messages and HTTP status codes.

WEB DEVELOPMENT:

1. Setup React Application:

- Create a React app in the client folder.
- Install required libraries
- Create required pages and components and add routes.

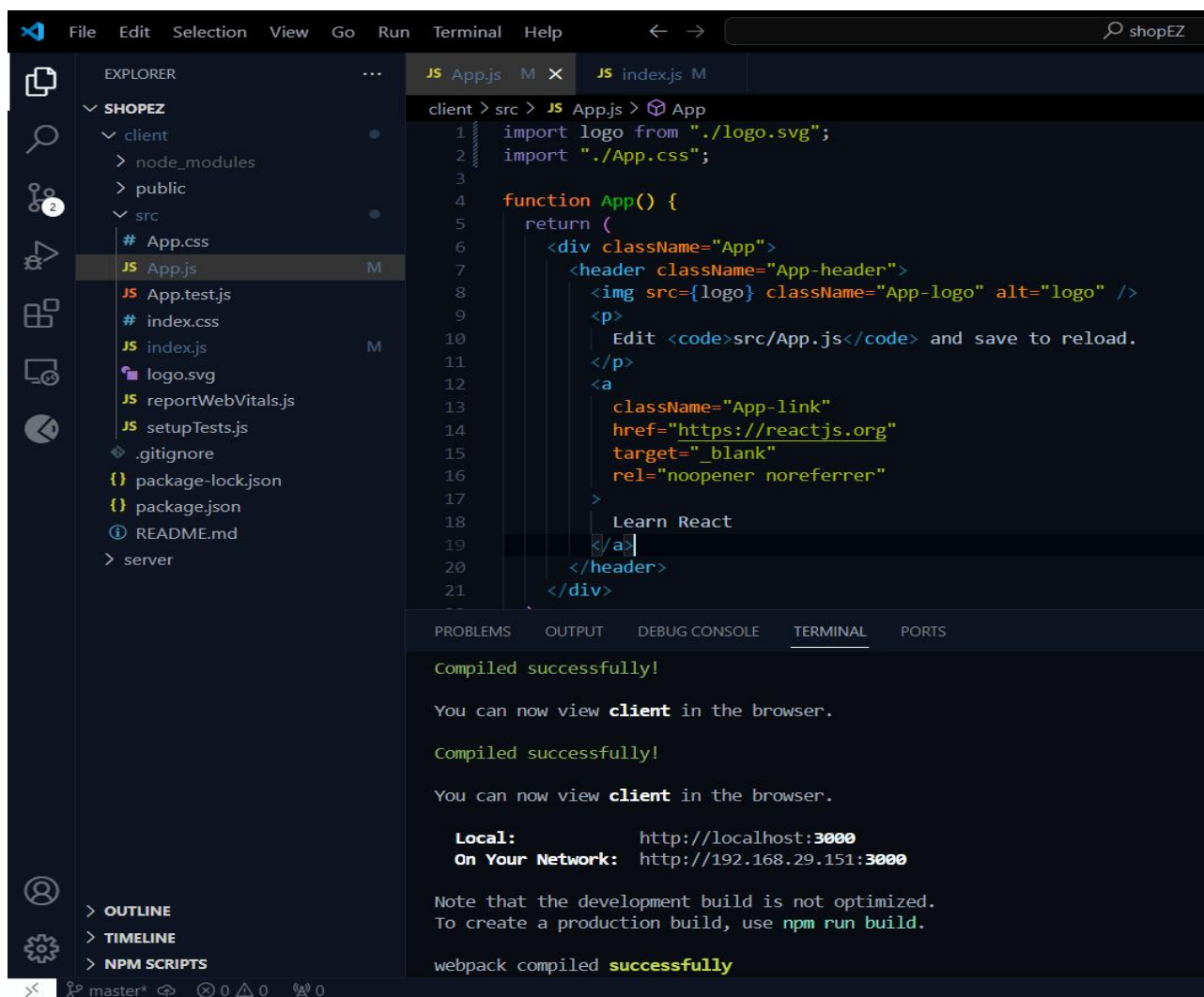
2.Design UI components:

- Create Components.
- Implement layout and styling.
- Add navigation.

3.Implement frontend logic:

- Integration with API endpoints.
- Implement data binding.

Reference Image:

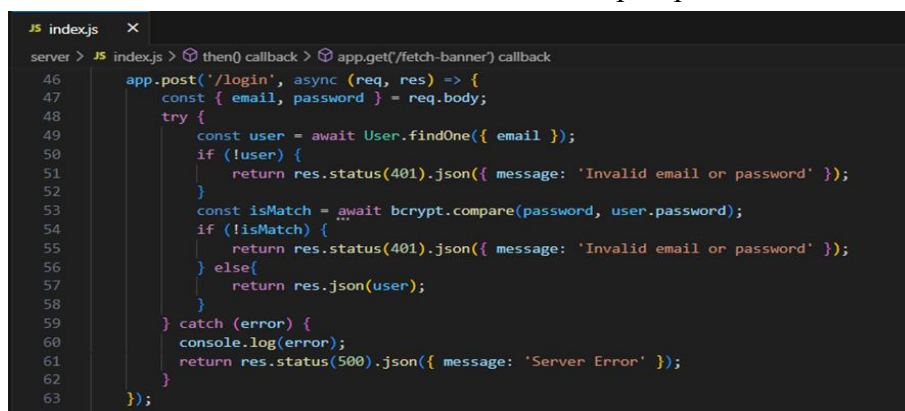


PROJECT IMPLEMENTATION & EXECUTION:

User Authentication:

Backend

Now, here we define the functions to handle http requests from the client for authentication.



Frontend Login:


```

JS GeneralContext.js U X
client > src > context > JS GeneralContext.js > [🔍] GeneralContextProvider > [🔍] register > [🔍] then() callback
46 const login = async () =>{
47   try{
48     const loginInputs = {email, password}
49     await axios.post('http://localhost:6001/login', loginInputs)
50     .then( async (res)=>{
51
52       localStorage.setItem('userId', res.data._id);
53       localStorage.setItem('userType', res.data.usertype);
54       localStorage.setItem('username', res.data.username);
55       localStorage.setItem('email', res.data.email);
56       if(res.data.usertype === 'customer'){
57         navigate('/');
58       } else if(res.data.usertype === 'admin'){
59         navigate('/admin');
60       }
61     }).catch((err) =>{
62       alert("login failed!!");
63       console.log(err);
64     });
65   }catch(err){
66     console.log(err);
67   }
68 }
69

```

Register:

```

JS GeneralContext.js U X
client > src > context > JS GeneralContext.js > [🔍] GeneralContextProvider > [🔍] logout
71 const inputs = {username, email, usertype, password};
72
73 const register = async () =>{
74   try{
75     await axios.post('http://localhost:6001/register', inputs)
76     .then( async (res)=>{
77       localStorage.setItem('userId', res.data._id);
78       localStorage.setItem('userType', res.data.usertype);
79       localStorage.setItem('username', res.data.username);
80       localStorage.setItem('email', res.data.email);
81
82       if(res.data.usertype === 'customer'){
83         navigate('/');
84       } else if(res.data.usertype === 'admin'){
85         navigate('/admin');
86       }
87     }).catch((err) =>{
88       alert("registration failed!!");
89       console.log(err);
90     });
91   }catch(err){
92     console.log(err);
93   }
94 }
95

```

logout:

```

GeneralContext.jsx U X
client > src > context > GeneralContext.jsx > [🔍] GeneralContextProvider > [🔍] login >
72
73 const logout = async () =>{
74
75   localStorage.clear();
76   for (let key in localStorage) {
77     if (localStorage.hasOwnProperty(key)) {
78       localStorage.removeItem(key);
79     }
80   }
81
82   navigate('/');
83 }
84
85

```

All Products (User):

In the home page, we'll fetch all the products available in the platform along with the filters.

Fetching products:

```
Products.jsx 2, U X
client > src > components > Products.jsx > Products > handleCategoryCheckBox
10  const [categories, setCategories] = useState([]);
11  const [products, setProducts] = useState([]);
12  const [visibleProducts, setVisibleProducts] = useState([]);
13
14  useEffect(()=>{
15    fetchData();
16  }, [])
17
18  const fetchData = async() =>{
19
20    await axios.get('http://localhost:6001/fetch-products').then(
21      (response)=>{
22        if(props.category === 'all'){
23          setProducts(response.data);
24          setVisibleProducts(response.data);
25        }else{
26          setProducts(response.data.filter(product=> product.category === props.category));
27          setVisibleProducts(response.data.filter(product=> product.category === props.category));
28        }
29      }
30    )
31    await axios.get('http://localhost:6001/fetch-categories').then(
32      (response)=>{
33        setCategories(response.data);
34      }
35    )
36  }
37
```

In the backend, we fetch all the products and then filter them on the client side.

```
JS index.js X
server > JS index.js > then() callback > app.get('/fetch-banner') callback
100
101  // fetch products
102
103  app.get('/fetch-products', async(req, res)=>{
104    try{
105      const products = await Product.find();
106      res.json(products);
107    }catch(err){
108      res.status(500).json({ message: 'Error occurred' });
109    }
110  })
111
```


Filtering products:

```
Products.jsx 2, U X
client > src > components > Products.jsx > [0] Products > [0] useEffect() callback
38 const [sortFilter, setSortFilter] = useState('popularity');
39 const [categoryFilter, setCategoryFilter] = useState([]);
40 const [genderFilter, setGenderFilter] = useState([]);
41
42 const handleCategoryCheckBox = (e) =>{
43   const value = e.target.value;
44   if(e.target.checked){
45     setCategoryFilter([...categoryFilter, value]);
46   }else{
47     setCategoryFilter(categoryFilter.filter(size=> size !== value));
48   }
49 }
50
51 const handleGenderCheckBox = (e) =>{
52   const value = e.target.value;
53   if(e.target.checked){
54     setGenderFilter([...genderFilter, value]);
55   }else{
56     setGenderFilter(genderFilter.filter(size=> size !== value));
57   }
58 }
59
60 const handleSortFilterChange = (e) =>{
61   const value = e.target.value;
62   setSortFilter(value);
63   if(value === 'low-price'){
64     setVisibleProducts(visibleProducts.sort((a,b)=> a.price - b.price))
65   } else if (value === 'high-price'){
66     setVisibleProducts(visibleProducts.sort((a,b)=> b.price - a.price))
67   }else if (value === 'discount'){
68     setVisibleProducts(visibleProducts.sort((a,b)=> b.discount - a.discount))
69   }
70 }
71
72 useEffect(()=>{
73
74   if (categoryFilter.length > 0 && genderFilter.length > 0){
75     setVisibleProducts(products.filter(product=> categoryFilter.includes(product.category) && genderFilter.includes(product.gender) ));
76   }else if(categoryFilter.length === 0 && genderFilter.length > 0){
77     setVisibleProducts(products.filter(product=> genderFilter.includes(product.gender) ));
78   } else if(categoryFilter.length > 0 && genderFilter.length === 0){
79     setVisibleProducts(products.filter(product=> categoryFilter.includes(product.category)));
80   }else{
81     setVisibleProducts(products);
82   }
83
84   [categoryFilter, genderFilter])
85
86 }
```

Add product to cart:

Here, we can add the product to the cart or can buy directly.

```
IndividualProduct.jsx 2, U X
client > src > pages > customer > IndividualProduct.jsx > [0] IndividualProduct
61 const buyNow = async() =>{
62   await axios.post('http://localhost:6001/buy-product',{userId, name, email, mobile, address,
63     pincode, title: productName, description: productDescription,
64     mainImg: productMainImg, size, quantity: productQuantity, price: productPrice,
65     discount: productDiscount, paymentMethod: paymentMethod, orderDate: new Date()}).then(
66     (response)=>{
67       alert('Order placed!!!');
68       navigate('/profile');
69     }
70   ).catch((err)=>{
71     alert("Order failed!!!");
72   })
73 }
74
75 const handleAddToCart = async() =>{
76   await axios.post('http://localhost:6001/add-to-cart', {userId, title: productName, description: productDescription,
77     mainImg: productMainImg, size, quantity: productQuantity, price: productPrice,
78     discount: productDiscount}).then(
79     (response)=>{
80       alert("product added to cart!!!");
81       navigate('/cart');
82     }
83   ).catch((err)=>{
84     alert("Operation failed!!!");
85   })
86 }
87 }
```

Backend : In the backend, if we want to buy, then with the address and payment method, we process

buying. If we need to add the product to the cart, then we add the product details along with the user Id to the cart collection.

Buy product:

```
JS index.js X
server > JS index.js > then() callback > app.post('/add-to-cart') callback
234 // buy product
235
236 app.post('/buy-product', async(req, res)=>{
237   const {userId, name, email, mobile, address, pincode, title,
238         description, mainImg, size, quantity, price,
239         discount, paymentMethod, orderDate} = req.body;
240   try{
241     const newOrder = new Orders({userId, name, email, mobile, address,
242                                pincode, title, description, mainImg, size, quantity, price,
243                                discount, paymentMethod, orderDate})
244     await newOrder.save();
245     res.json({message: 'order placed'});
246   }catch(err){
247     res.status(500).json({message: "Error occured"});
248   }
249 })
```

Add product to cart:

```
JS index.js X
server > JS index.js > then() callback > app.put('/increase-cart-quantity') callback
301 // add cart item
302
303 app.post('/add-to-cart', async(req, res)=>{
304
305   const {userId, title, description, mainImg, size, quantity, price, discount} = req.body
306   try{
307     const item = new Cart({userId, title, description, mainImg, size, quantity, price, discount});
308     await item.save();
309     res.json({message: 'Added to cart'});
310   }catch(err){
311     res.status(500).json({message: "Error occured"});
312   }
313 })
314
```

Order products:

Now, from the cart, let's place the order

Frontend

```
Cart.jsx 3, U X
client > src > pages > customer > Cart.jsx > Cart
83 const [name, setName] = useState('');
84 const [mobile, setMobile] = useState('');
85 const [email, setEmail] = useState('');
86 const [address, setAddress] = useState('');
87 const [pincode, setPincode] = useState('');
88 const [paymentMethod, setPaymentMethod] = useState('');
89
90 const userId = localStorage.getItem('userId');
91 const placeOrder = async() =>{
92   if(cartItems.length > 0){
93     await axios.post('http://localhost:6001/place-cart-order', {userId, name, mobile,
94                                                                email, address, pincode, paymentMethod, orderDate: new Date()}).then(
95       (response)=>{
96         alert('Order placed!!');
97         setName('');
98         setMobile('');
99         setEmail('');
100        setAddress('');
101        setPincode('');
102        setPaymentMethod('');
103        navigate('/profile');
104      }
105    )
106  }
107 }
```

Backend

In the backend, on receiving the request from the client, we then place the order for the products in the cart with the specific user Id.

```
JS index.js X
server > JS index.js > then() callback
359
360 // Order from cart
361
362 app.post('/place-cart-order', async(req, res)=>{
363   const {userId, name, mobile, email, address, pincode, paymentMethod, orderDate} = req.body;
364   try{
365     const cartItems = await Cart.find({userId});
366     cartItems.map(async (item)=>{
367       const newOrder = new Orders({userId, name, email, mobile, address,
368         pincode, title: item.title, description: item.description,
369         mainImg: item.mainImg, size:item.size, quantity: item.quantity,
370         price: item.price, discount: item.discount, paymentMethod, orderDate});
371       await newOrder.save();
372       await Cart.deleteOne({_id: item._id})
373     })
374     res.json({message: 'Order placed'});
375   }catch(err){
376     res.status(500).json({message: "Error occured"});
377   }
378 }
379 })
380
```

Add new product:

Here, in the admin dashboard, we will add a new product.

o Frontend:

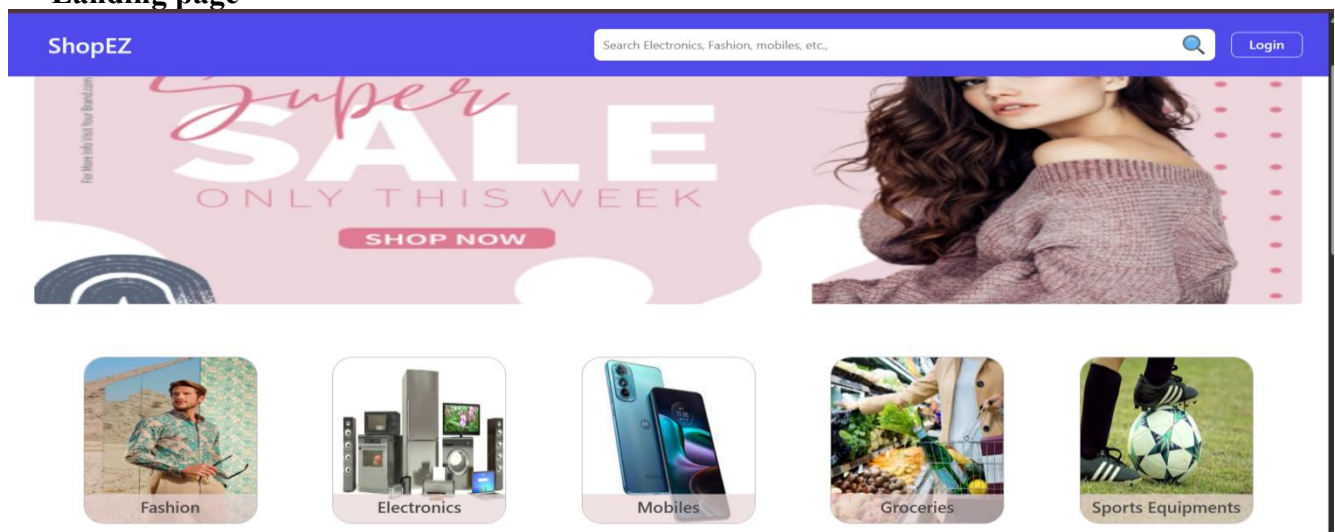
```
NewProduct.jsx X
client > src > pages > admin > NewProduct.jsx > NewProduct
47
48 const handleNewProduct = async() =>{
49   await axios.post('http://localhost:6001/add-new-product', {productName, productDescription, productMainImg,
50     productCarousel: [productCarouselImg1, productCarouselImg2, productCarouselImg3], productSizes,
51     productGender, productCategory, productNewCategory, productPrice, productDiscount}).then(
52     (response)=>{
53       alert("product added");
54       setProductName('');
55       setProductDescription('');
56       setProductMainImg('');
57       setProductCarouselImg1('');
58       setProductCarouselImg2('');
59       setProductCarouselImg3('');
60       setProductSizes([]);
61       setProductGender('');
62       setProductCategory('');
63       setProductNewCategory('');
64       setProductPrice(0);
65       setProductDiscount(0);
66     }
67     navigate('/all-products');
68   }
69 )
70 }
```


Backend:

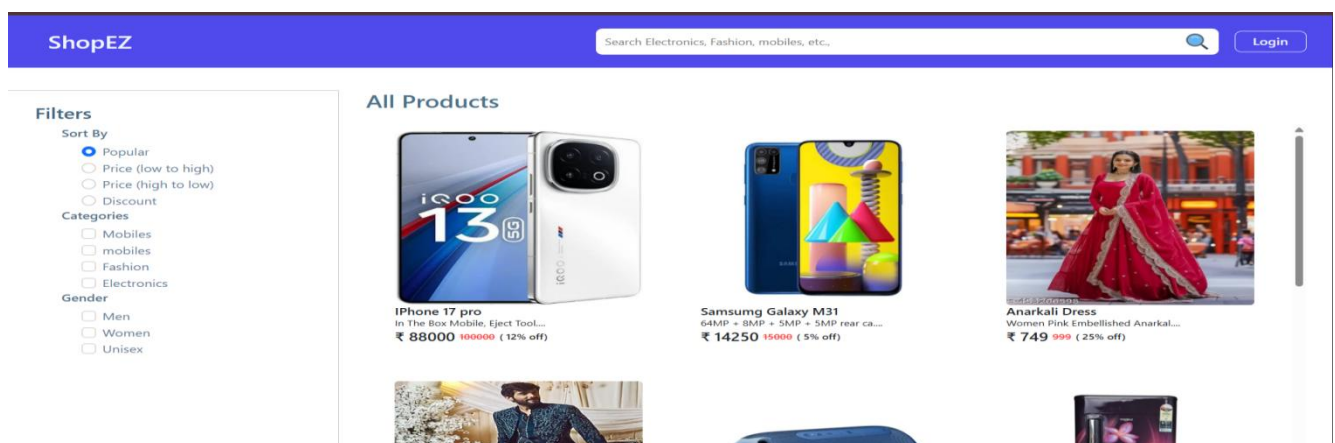
```
JS index.js X
server > JS index.js > then() callback > app.put('/update-product/id') callback
143
144 // Add new product
145
146 app.post('/add-new-product', async(req, res)=>{
147   const {productName, productDescription, productMainImg, productCarousel,
148         productSizes, productGender, productCategory, productNewCategory,
149         productPrice, productDiscount} = req.body;
150   try{
151     if(productCategory === 'new category'){
152       const admin = await Admin.findOne();
153       admin.categories.push(productNewCategory);
154       await admin.save();
155       const newProduct = new Product({title: productName, description: productDescription,
156                                       mainImg: productMainImg, carousel: productCarousel, category: productNewCategory,
157                                       sizes: productSizes, gender: productGender, price: productPrice, discount: productDiscount});
158       await newProduct.save();
159     } else{
160       const newProduct = new Product({title: productName, description: productDescription,
161                                       mainImg: productMainImg, carousel: productCarousel, category: productCategory,
162                                       sizes: productSizes, gender: productGender, price: productPrice, discount: productDiscount});
163       await newProduct.save();
164     }
165     res.json({message: "product added!!"});
166   }catch(err){
167     res.status(500).json({message: "Error occured"});
168   }
169 })
170
```

Demo UI images:

· Landing page



· Products



· Authentication

ShopEZ

Search Electronics, Fashion, mobiles, etc.,

Login

Register

Username

Email address

Password

User type

▼

User type

Admin

Customer

Already registered? Login

· User Profile

ShopEZ

Search Electronics, Fashion, mobiles, etc.,

Sohith Kumar

0

Username: Sohith Kumar

Email: sohithkumarn@gmail.com

Orders: 4

Logout

Orders



Vivo V50e

From close-ups to full frames, the V50e's Multifocal Pro Portrait gives you three classic portrait focal lengths to bring your vision to life.

Size: Quantity: 1 Price: ₹ 31319 Payment method: cod

Address: ramalayam street Pincode: 522408 Ordered on: 2025-06-24

Order status: Order placed



Whirlpool Fridge 180L

✓ Whirlpool Fridge ✓ 180 Liters Capacity ✓ 1 Star Rating ✓ Purple Gloria ✓ 10 Years Warranty

Size: Quantity: 1 Price: ₹ 41479 Payment method: cod

Address: ramalayam street Pincode: 522408 Ordered on: 2025-06-24

Order status: delivered

· Cart

ShopEZ

Search Electronics, Fashion, mobiles, etc.,

Sohith Kumar

1



boAt Stone 358pro

1 Unit Speaker, 1 Unit Cable, 1 Unit User Manual, 1 Unit Warranty Card, 1 Unit Catalogue, 1 Unit FAQ Card Read more at: <https://www.boat-lifestyle.com/products/boat-stone-358-pro-portable-bluetooth-speaker-with-14w-boat-signature-sound-rgb-led-design-12-hours-playback>

Size: Quantity: 1

Price: ₹ 1679

Remove

Price Details

Total MRP:

₹ 1999

Discount on MRP:

- ₹ 319

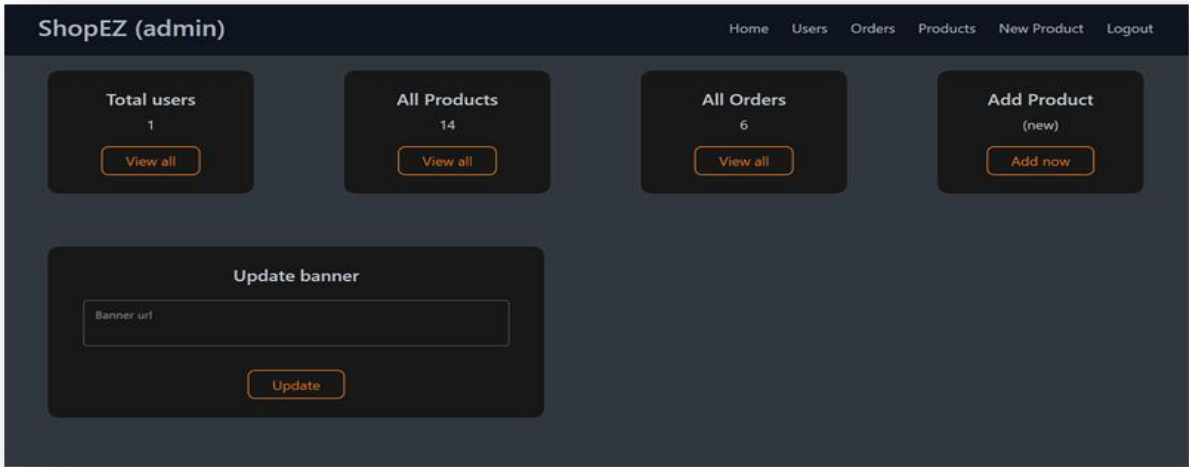
Delivery Charges:

+ ₹ 0

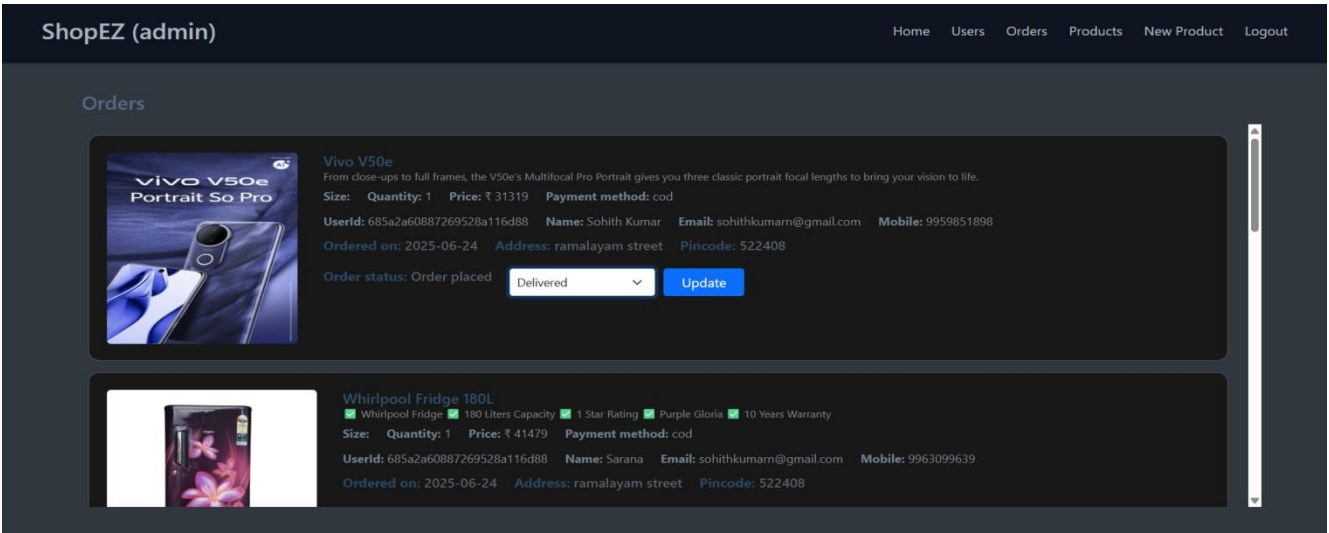
Final Price: ₹ 1680

Place order

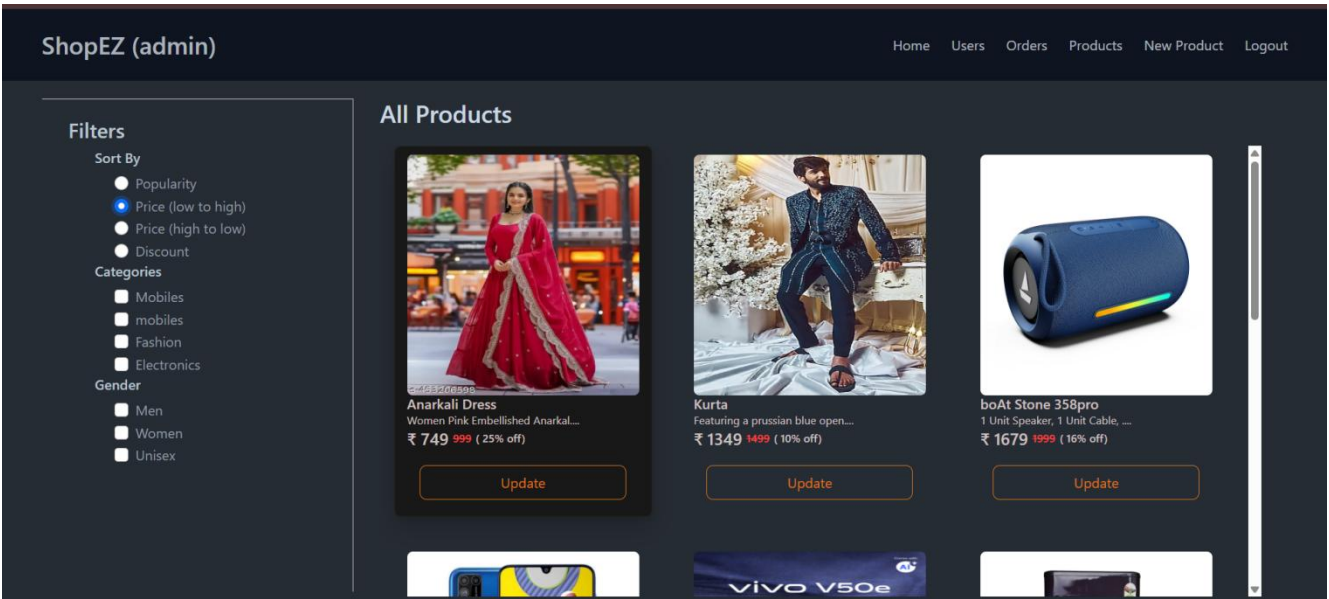
Admin dashboard



All Orders



All Products



New Product Page

ShopEZ (admin) Home Users Orders Products New Product Logout

New Product

Product name Product Description

Thumbnail img url

Add on img1 url Add on img2 url Add on img3 url

Available Size

☐ S ☐ M ☐ L ☐ XL

Gender

Add product

Conclusion:

Shopez empowers individuals and businesses to take control of their shopping and selling experience, one order at a time. By offering a smart, secure, and user-friendly e-commerce platform, we aim to unlock the full potential of online retail and create seamless connections between buyers and sellers in a growing digital marketplace.

GitHub Link:

https://github.com/SohithC5/ShopEZ_E-Commerce-App-MERN

Project Demo Link :

<https://drive.google.com/file/d/1Q0q8CSI5AnsXR7tztt55K35pyv67VR82/view?usp=sharing>

**** Happy Coding ****